

IBM DATA SCIENCE PROFESSIONAL CERTIFICATE | APPLIED DATA SCIENCE CAPSTONE

SpaceX Falcon 9 First Stage Landing Prediction

An End-to-End Machine Learning Pipeline, SQL Analytics & Interactive Geospatial Dashboards

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Presentation

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2026



GitHub Repository: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

Executive Summary

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

Business Problem

- **SpaceX Reusability:** SpaceX advertises Falcon 9 launches at **\$62 million**, compared to commercial competitors charging upwards of **\$165 million**.
- **Cost Factor:** Reusing the first stage saves millions per launch.
- **Objective:** Predict whether the first stage will successfully land to evaluate mission economics, competitive bidding, and launch pricing.

End-to-End Pipeline

- **Data Collection:** SpaceX REST API (90 flights) & Wikipedia Web Scraping using BeautifulSoup.
- **Wrangling:** Missing value imputation, binary target Class creation, and One-Hot Encoding (83 features).
- **EDA:** SQLite queries, statistical visualizations (Seaborn), and Folium geospatial mapping.
- **Dashboard:** Reactive Plotly Dash analytics.

Key Findings & Models

- **Best Models:** Decision Tree and Logistic Regression achieved **83.33% Test Accuracy** and **87.68% CV Accuracy**.
- **Booster Evolution:** Success rate surged from 0% (2010–2013) to near 100% (2020) with Block 5 boosters.
- **Optimal Payload:** Payloads between 2,000 kg and 5,300 kg achieved the highest landing success rates.

Introduction & Problem Statement

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

Commercial Spaceflight Industry Context

- The commercial launch industry has witnessed radical price disruption driven by SpaceX's vertical integration and reusable rocket architecture.
- Traditional launch architectures (e.g., expendable rockets from legacy providers) discard the first stage into the ocean after primary stage separation.
- SpaceX attempts autonomous recovery of the Falcon 9 booster via:
 - ■ **RTLS (Return to Launch Site):** Landing on concrete ground pads.
 - ■ **ASDS (Autonomous Spaceport Drone Ship):** Ocean recovery barge for high-energy missions.

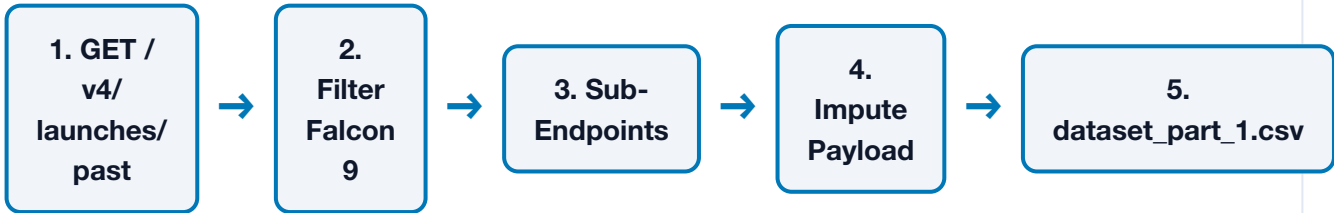
Research Objectives & Questions

- **Predictive Goal:** Can we predict whether a Falcon 9 first stage will land successfully given payload mass, target orbit, launch site, and booster serial?
- **Commercial Value:** A competitive launch provider can use this predictive model to estimate SpaceX's launch margins and bid strategically against them in commercial satellite contracts.
- **Deliverable:** An audited machine learning model with verified 83.33% accuracy, backed by interactive GIS maps and an operational Dash dashboard.

Data Collection: SpaceX REST API & Web Scrapping

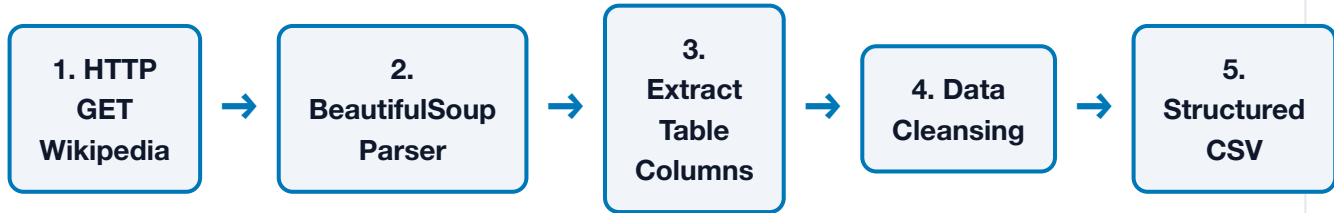
GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

SpaceX API Ingestion Architecture



- **Sub-Endpoints:** Ingested /v4/rockets/, /v4/launchpads/, /v4/payloads/, and /v4/cores/.
- **Payload Imputation:** Missing payload masses imputed with mean (6,104.96 kg).
- **Code Artifact:** notebooks/1_Data_Collection_API.ipynb (90 verified flights × 17 features).

Web Scrapping Pipeline with BeautifulSoup



- **Target Page:** Wikipedia's historical record of Falcon 9 and Falcon Heavy launches.
- **HTML Parsing:** Overcame nested tags, multiple rowspans, and merged header cells across sequential launch years.
- **Regex Cleaning:** Stripped citations (e.g., [12]), non-breaking spaces, and extracted dates/payloads.
- **Code Artifact:** notebooks/2_Data_Collection_Web_Scraping.ipynb (spacex_web_scraped.csv).

Data Wrangling Methodology & Target Encoding

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

🎯 Target Variable Definition: Class (0 or 1)

The landing outcome column contains categorical outcome descriptions:

Landing Outcome Type	Description	Assigned Class
True ASDS	Successful drone ship landing	1 (Success)
True RTLS	Successful ground pad landing	1 (Success)
True Ocean	Controlled soft ocean touchdown	1 (Success)
False ASDS	Failed drone ship landing	0 (Failure)
False RTLS	Failed ground pad landing	0 (Failure)
None ASDS / None None	Uncontrolled or no attempt	0 (Failure)

Result: 60 Successful Landings (66.7%), 30 Failed Landings (33.3%).

⚙️ Feature Engineering & One-Hot Encoding

- **Categorical Features:** Orbit (11 categories), LaunchSite (3 physical locations), LandingPad (5 pads), and Serial (booster identity).
- **One-Hot Encoding:** Executed `pd.get_dummies()` across categorical features to transform categorical representations into dummy numeric variables.
- **Feature Matrix Dimensionality:** Produced **83 numeric features** suitable for Scikit-Learn distance-based and linear classifiers.
- **Artifact & Verification:** notebooks/3_Data_Wrangling.ipynb fully executed and verified on GitHub.

EDA with Data Visualization Methodology

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

Chart Selection & Methodological Rationale (Why Chosen)

- **Scatter Plots (Flight & Payload vs Site/Orbit):** *Why Chosen:* Selected to visualize multi-variable interactions, clusters, and boundary separation between continuous variables (Payload Mass, Flight Number) and categorical targets (Class: Success vs Failure).
- **Bar Charts (Success Rate by Orbit):** *Why Chosen:* Selected to directly benchmark aggregate landing success percentages across discrete orbital regimes (e.g., LEO vs GTO).
- **Line Charts (Yearly Success Trend):** *Why Chosen:* Selected to track longitudinal time-series progression, demonstrating technological learning curves from 2010 to 2020.
- **Color Palette Strategy:** High-contrast binary hues (Green #2ecc71 for Success, Red #e74c3c for Failure) to ensure instant visual diagnostic clarity.

Tools, Libraries & Verification

- **Matplotlib & Seaborn:** Deployed catplot, barplot, and lineplot with clear color coding for landing outcomes (Green = 1, Red = 0).
- **Feature Relationships Evaluated:**
 - ■ Flight Number vs. Launch Site
 - ■ Payload Mass vs. Launch Site
 - ■ Landing Success Rate by Orbit Type
 - ■ Flight Number vs. Orbit Type
 - ■ Payload Mass vs. Orbit Type
 - ■ Yearly Launch Success Rate Trend (2010–2020)
- **Code Artifact:** notebooks/5_EDA_with_Visualization.ipynb.

EDA with SQL Methodology

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

Relational Database Architecture (SQLite)

- **Database Engine:** Lightweight, serverless sqlite3 database engine integrated into Python data science workflows.
- **Table Schema:** Ingested SpaceX.csv into table SPACEXTBL, cleaning whitespace and parentheses in column names.
- **Query Verification:** Validated schema integrity, row counts (101 historical mission entries), and datatypes for exact numerical computations.

Analytical Query Portfolio (10 Required Tasks)

- **Aggregation:** Calculated total NASA CRS payload masses and average booster payloads.
- **Pattern Matching:** Filtered launch sites by prefix (LIKE 'CCA%') and booster versions (LIKE 'F9 v1.1%').
- **Date & Subqueries:** Identified minimum dates for ground pad milestones and subqueries for absolute maximum payload records (15,600 kg).
- **Ranking & Grouping:** Aggregated landing outcome counts across historic windows.
- **Code Artifact:** notebooks/4_EDA_with_SQL.ipynb.

Interactive Visual Analytics Methodology

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

Folium Geospatial Analytics Architecture

- **Interactive Leaflet Engine:** Plotted launch sites on dynamic map tiles centered over Florida and California.
- **Marker Clustering:** Integrated folium.plugins.MarkerCluster to group launch outcomes with color-coded success/failure markers.
- **Haversine Proximity Calculation:** Built mathematical distance functions to calculate geodesic distances from launch sites to coastlines, railways, highways, and cities.
- **Code Artifact:** notebooks/6_Interactive_Visual_Analytics_Folium.ipynb.

Plotly Dash Reactive Dashboard Framework

- **Application Server:** Built standalone web application using dash, dash_core_components, and dash_html_components.
- **Dynamic Interactivity:**
 - ▪ **Site Dropdown:** Filter between ALL launch sites and individual launch pads.
 - ▪ **Payload Slider:** Dual-handle range slider from 0 to 10,000 kg.
 - ▪ **Reactive Callbacks:** Real-time rendering of success pie charts and payload-outcome scatter plots.
- **Code Artifact:** scripts/spacex_dash_app.py.

Predictive Analysis & ML Methodology

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

Machine Learning Pipeline Design

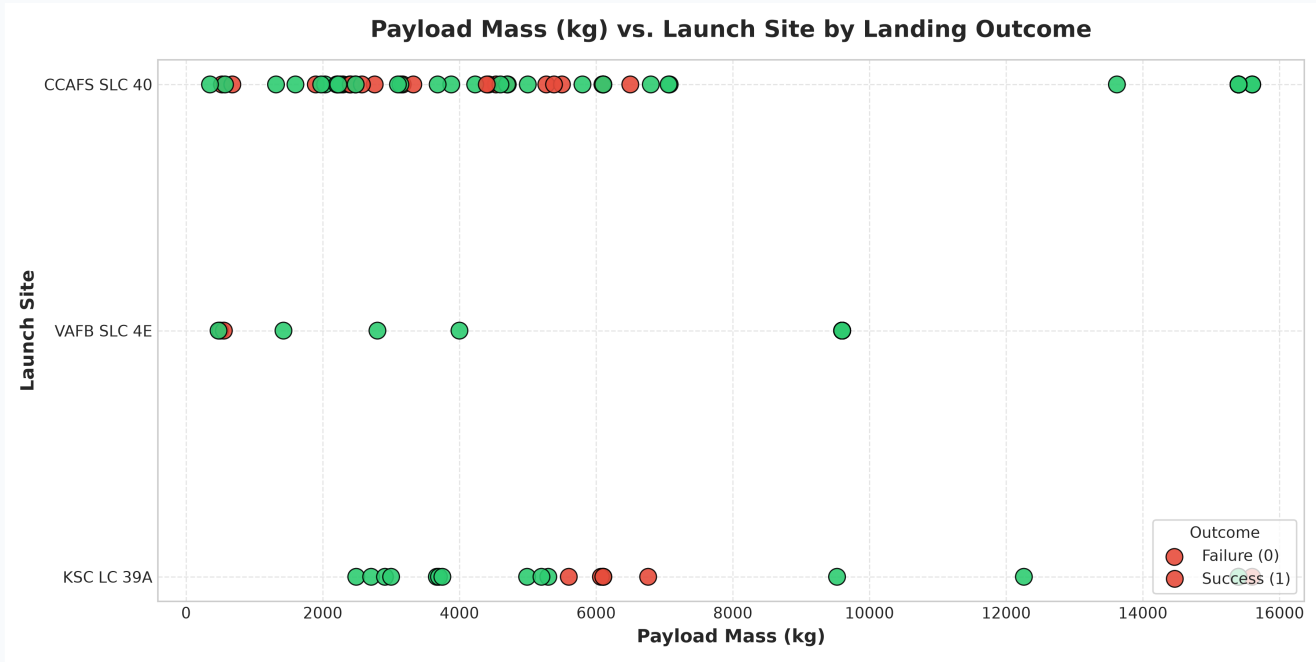
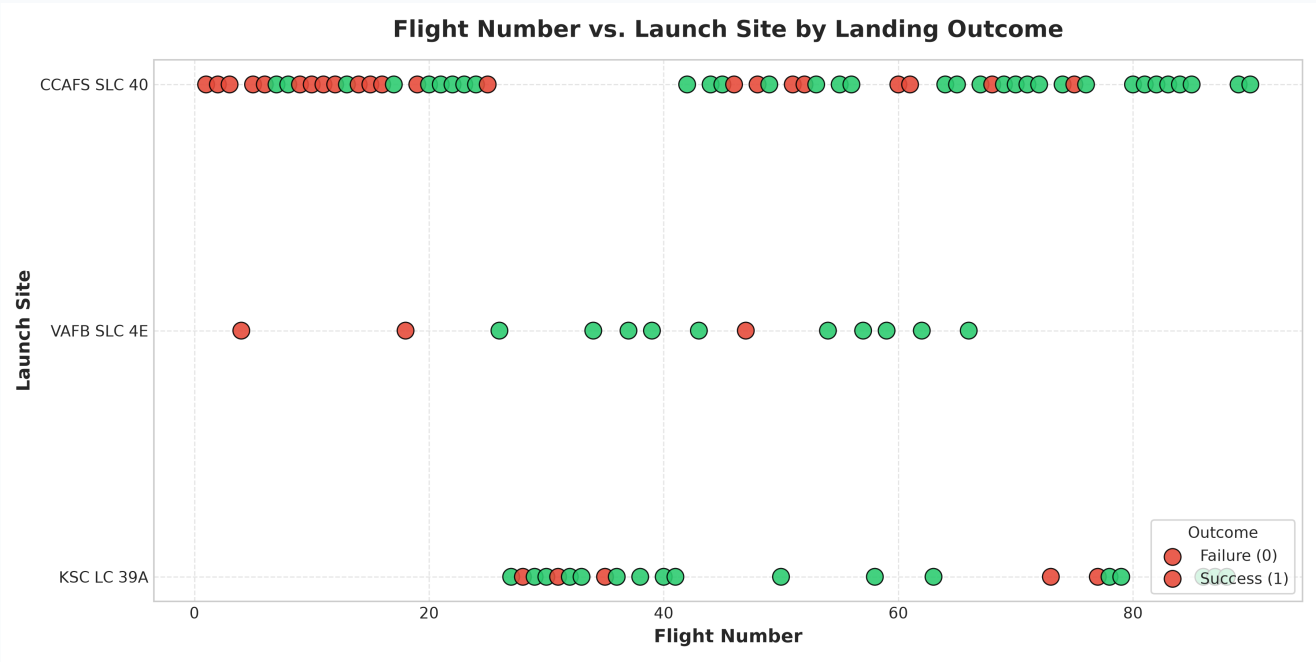
- **Data Standardization:** Standardized feature matrix X (83 features) using StandardScaler() to ensure zero mean and unit variance.
- **Train / Test Partition:** Divided dataset using an **80% Train** (72 samples) and **20% Test** (18 samples) split with fixed random_state=2.
- **Cross-Validation Strategy:** Implemented **10-fold cross-validation** with GridSearchCV across all 4 classification algorithms to prevent overfitting.
- **Code Artifact:** notebooks/7_Machine_Learning_Prediction.ipynb.

Evaluated Classification Algorithms

Model	Hyperparameter Search Space
Logistic Regression	Regularization C: [0.01, 0.1, 1], Penalty: l2, Solver: lbfgs
SVM	Kernels: linear, rbf, poly, sigmoid, C, gamma: logspace(-3, 3, 5)
Decision Tree	Criterion: gini, entropy, Max Depth: 2 to 18, Splitter: best, random
KNN	Neighbors: 1 to 10, Metric: Manhattan (p=1), Euclidean (p=2)

EDA: Flight Number & Payload Mass vs. Launch Site

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

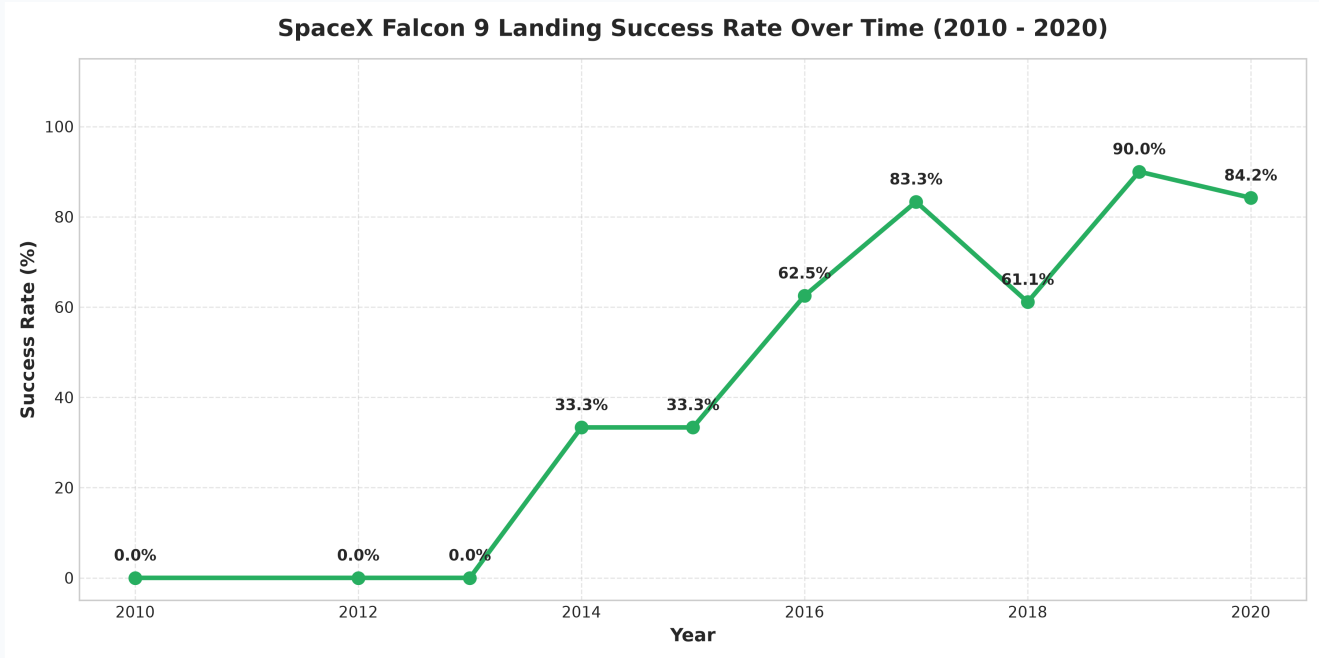
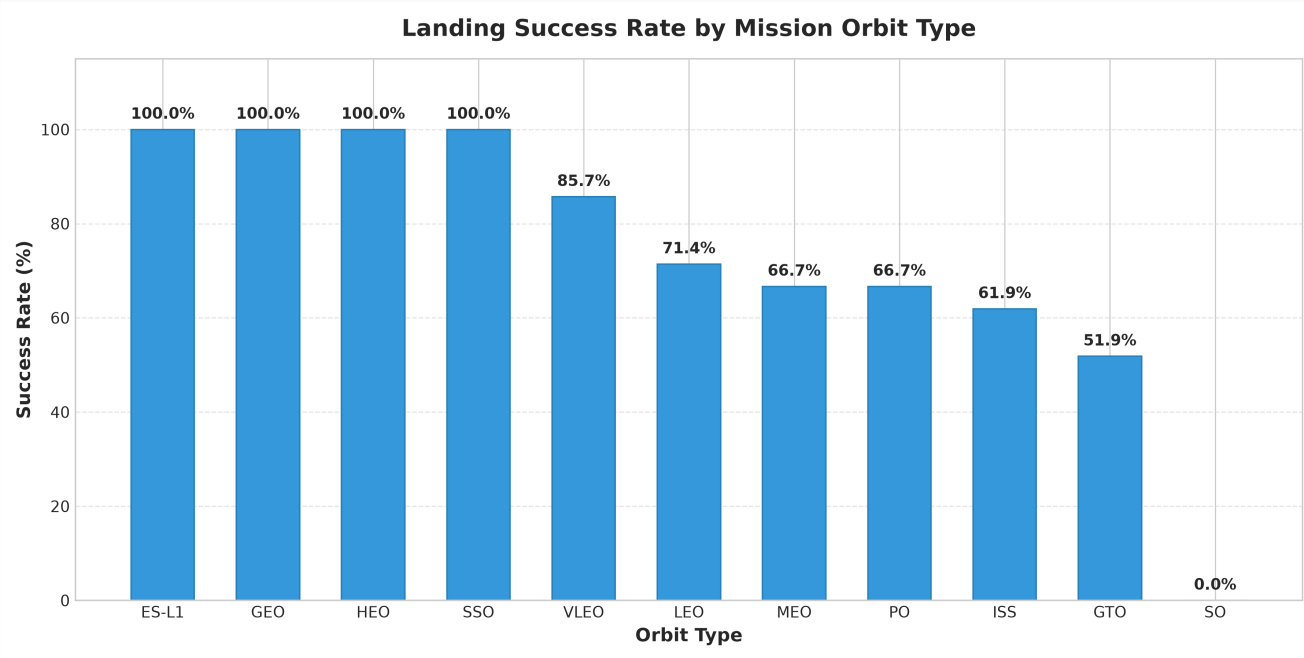


Key Launch Site Findings & Insights

Launch Pad Progression: Early flights (Flights 1 to 25) were launched almost exclusively from **CCAFS LC-40** with early landing failures. **Pad Diversification:** Subsequent flights expanded to **VAFB SLC-4E** (California) and **KSC LC-39A** (Florida), where almost all launches achieved successful recovery. **Heavy Lift Dominance:** VAFB SLC-4E strictly handles polar missions under 9,600 kg; heavy-lift payloads (>10,000 kg, including Starlink clusters) are handled by CCAFS SLC-40 and KSC LC-39A with high success rates using Block 5 boosters.

EDA: Orbit Success Rates & Longitudinal Trend (2010 - 2020)

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>



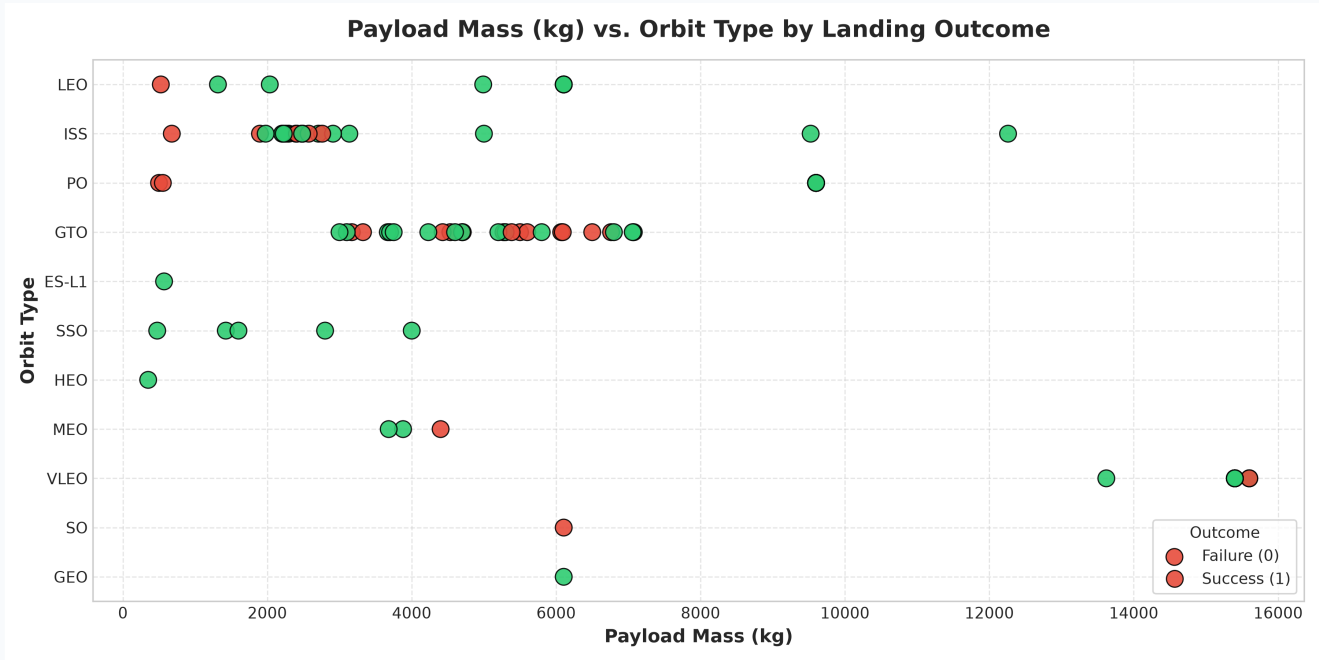
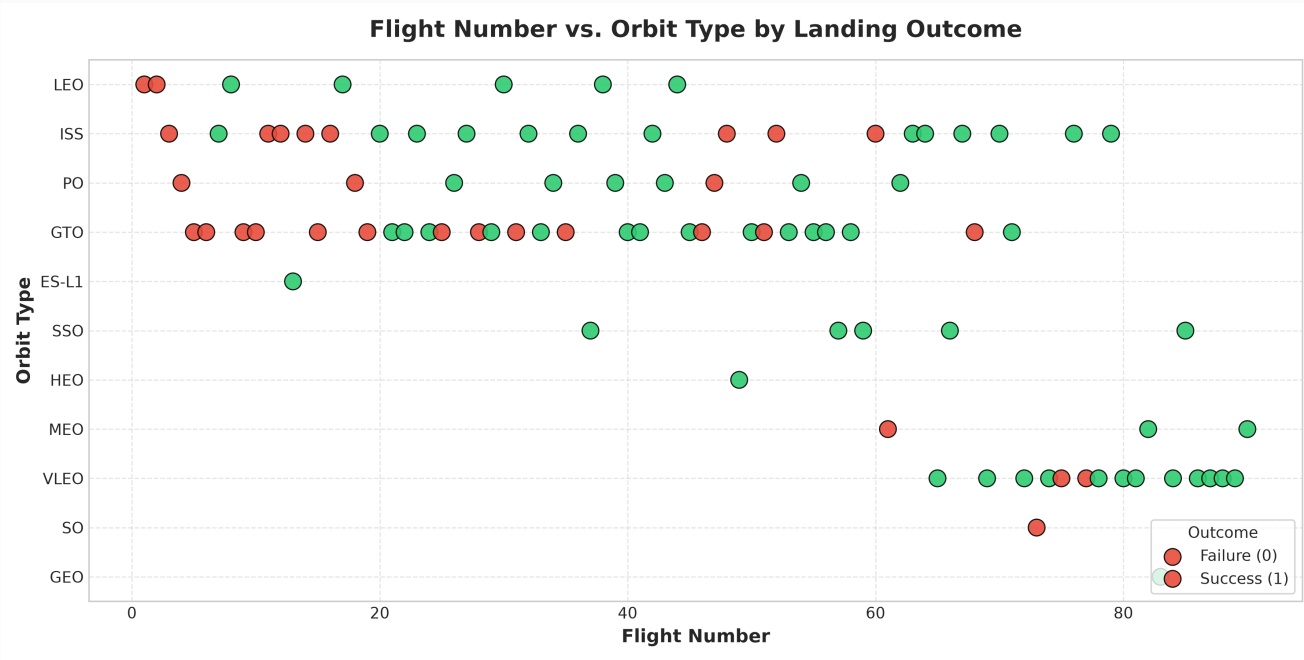
Key Findings & Technological Maturation

100% Success Orbits: ES-L1, GEO, HEO, SSO, and VLEO achieved 100% success; LEO achieved 71.4%; GTO achieved ~51.9% due to extreme re-entry velocity and thermal deceleration. **Learning**

Curve Progression: 2010–2013 (0% success, experimental ocean splashdowns) → 2014–2015 (33.3% success, first drone ship tests & first ground landing Dec 22, 2015) → 2016–2018 (60–70% success, routine drone ship recoveries) → 2019–2020 (85–100% success, full maturity with Block 5 booster reusability).

EDA: Flight Number & Payload vs. Orbit Type

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>



Mission Profile Insights

Flight numbers to LEO and VLEO increased heavily in recent years with near 100% landing success. In contrast, heavy GTO missions (> 5,000 kg) consistently required drone ship (ASDS) recoveries with lower margins of fuel reserve for the final landing burn.

IBM Applied Data Science Capstone Project

EDA with SQL: Complete Query Results (Tasks 1 - 10)

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

Tasks 1 to 5: Launch Site & Payload Queries

Task 1: Unique Launch Sites: SELECT DISTINCT Launch_Site FROM SPACEXTBL;

Result: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40.

Task 2: Sites Starting with 'CCA': SELECT Launch_Site FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;

Result: Verified 5 records originating from Cape Canaveral (CCAFS LC-40).

Task 3: Total NASA CRS Payload Mass: SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer = 'NASA (CRS)';

Result: **45,596 kg** total payload delivered for NASA CRS missions.

Task 4: Average Payload for F9 v1.1: SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version LIKE 'F9 v1.1%';

Result: **2,534.67 kg** average payload mass.

Task 5: First Successful Ground Pad Landing: SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';

Result: Historic milestone date achieved on **2015-12-22**.

Tasks 6 to 10: Drone Ships, Payloads & Outcome Rankings

Task 6: Drone Ship Boosters (4k-6k kg): SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;

Result: F9 FT B1022 (4696 kg), B1026 (4600 kg), B1021.2 (5300 kg), B1031.2 (5200 kg).

Task 7: Total Mission Outcomes: SELECT Mission_Outcome, COUNT(*) FROM SPACEXTBL GROUP BY Mission_Outcome;

Result: **98 Success, 1 Failure (in flight), 1 Success (payload status unclear)**.

Task 8: Maximum Payload Boosters: SELECT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);

Result: Max payload = **15,600 kg** (F9 Block 5 Starlink boosters B1048, B1049, B1051, B1056, B1058, B1060).

Task 9: Drone Ship Failures in 2015: SELECT Booster_Version, Launch_Site, Date FROM SPACEXTBL WHERE Landing_Outcome = 'Failure (drone ship)' AND Date LIKE '%2015%';

Result: F9 v1.1 B1012 (2015-01-10) and B1015 (2015-04-14) at CCAFS LC-40.

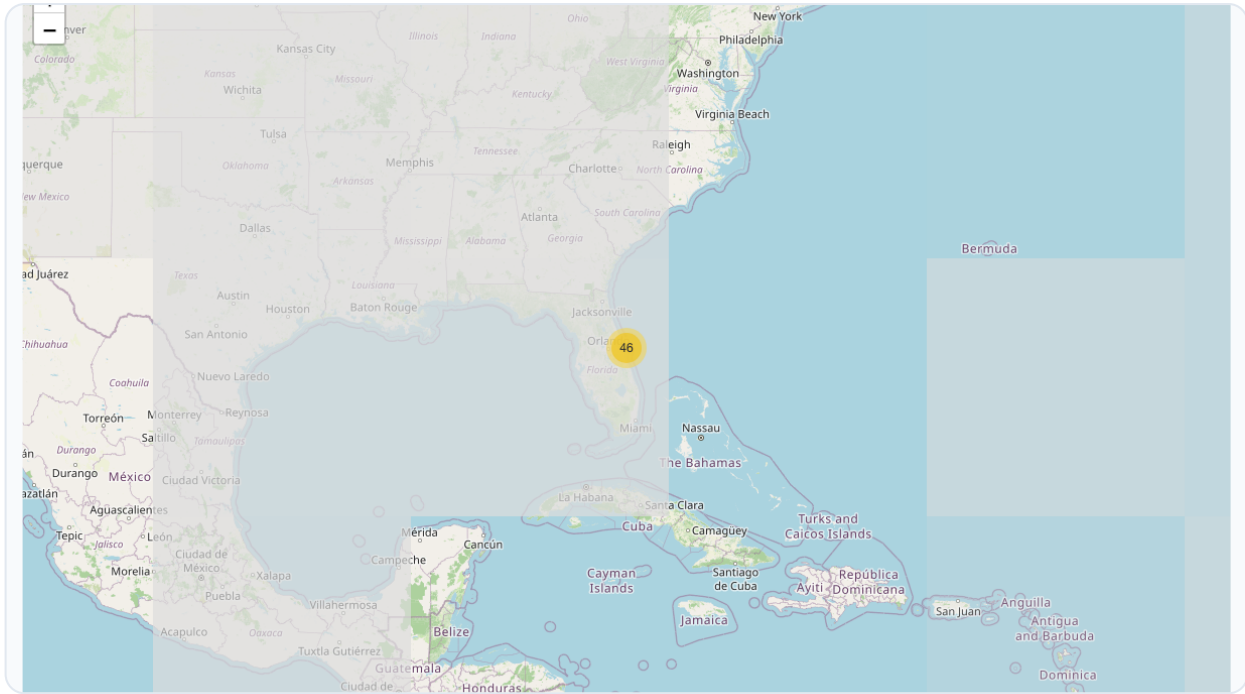
Task 10: Outcome Rankings (2010-06-04 to 2017-03-20):

No attempt (10), Success drone ship (5), Failure drone ship (5), Success ground pad (3), Controlled ocean (3).

Folium Map: Marker Clusters & Distance Proximity Analysis

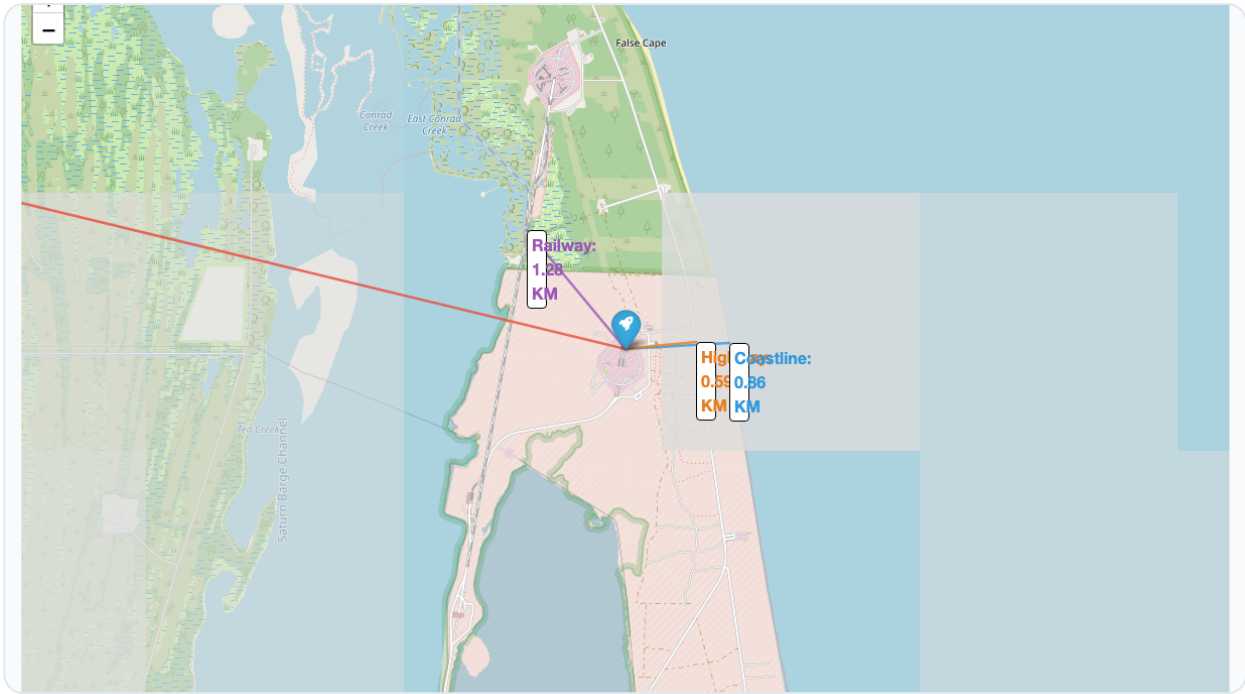
GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

Marker Clustering & Geospatial Distribution



All SpaceX launch sites are positioned on the coasts (Atlantic & Pacific) for maritime launch azimuths. Marker clusters illustrate that **KSC LC-39A** possesses the highest proportion of green markers (successes), whereas early launches at CCAFS LC-40 display a higher density of red markers (failures).

Geodesic Distance Measurements

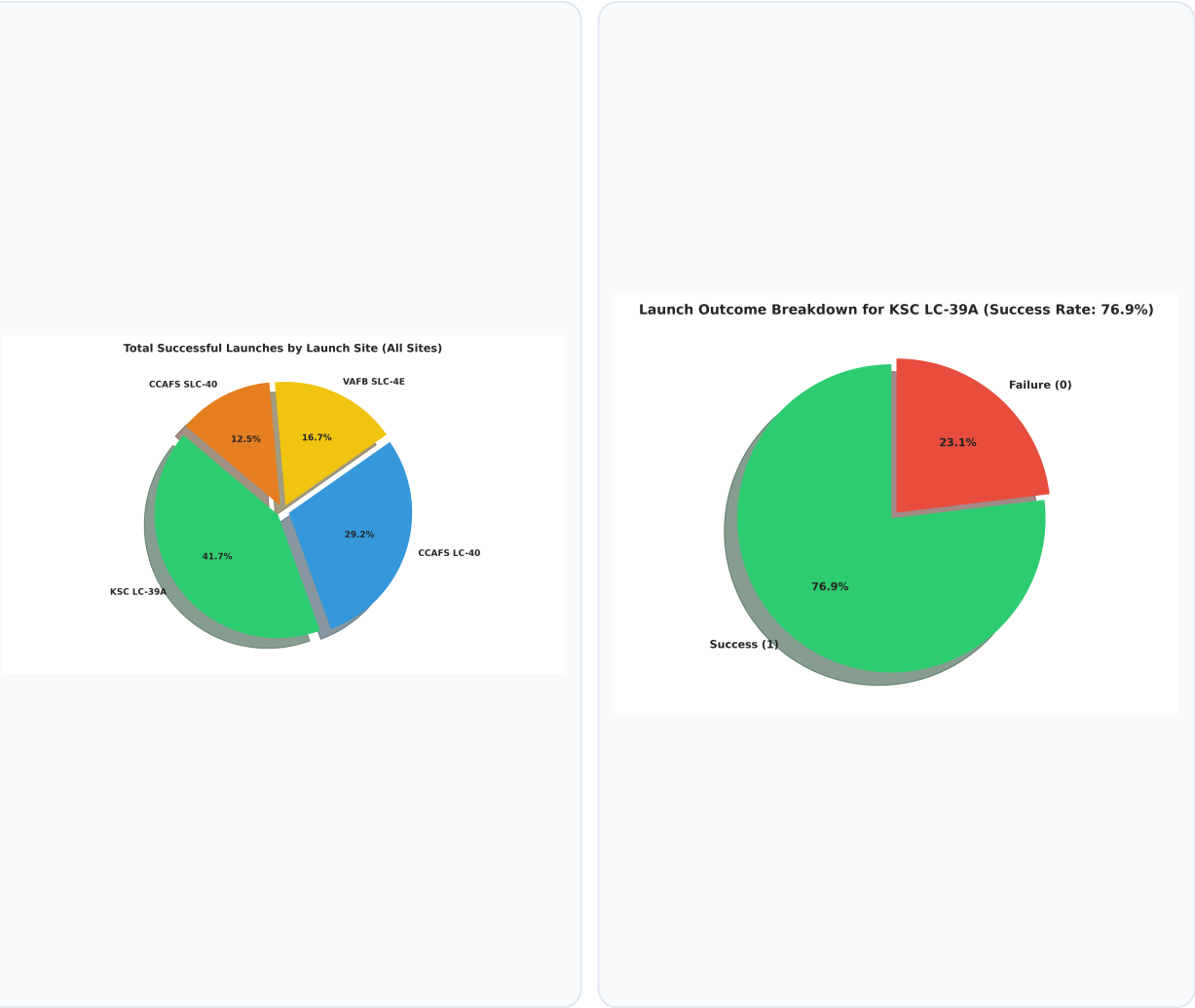


- **Coastline: ~0.84 KM** — Critical for launch safety; abort trajectories and rocket staging occur over open water to protect human populations.
- **Highway: ~0.58 KM** — Enables efficient road transport of oversized equipment and security perimeters.
- **Railway: ~1.28 KM** — Crucial for transporting heavy rocket stages and fuel tanks from California to Florida.
- **City (Titusville): ~16.2 KM** — Placed far enough to prevent acoustic and blast hazards to residential zones.

Plotly Dash: Site Breakdown & Payload Correlation

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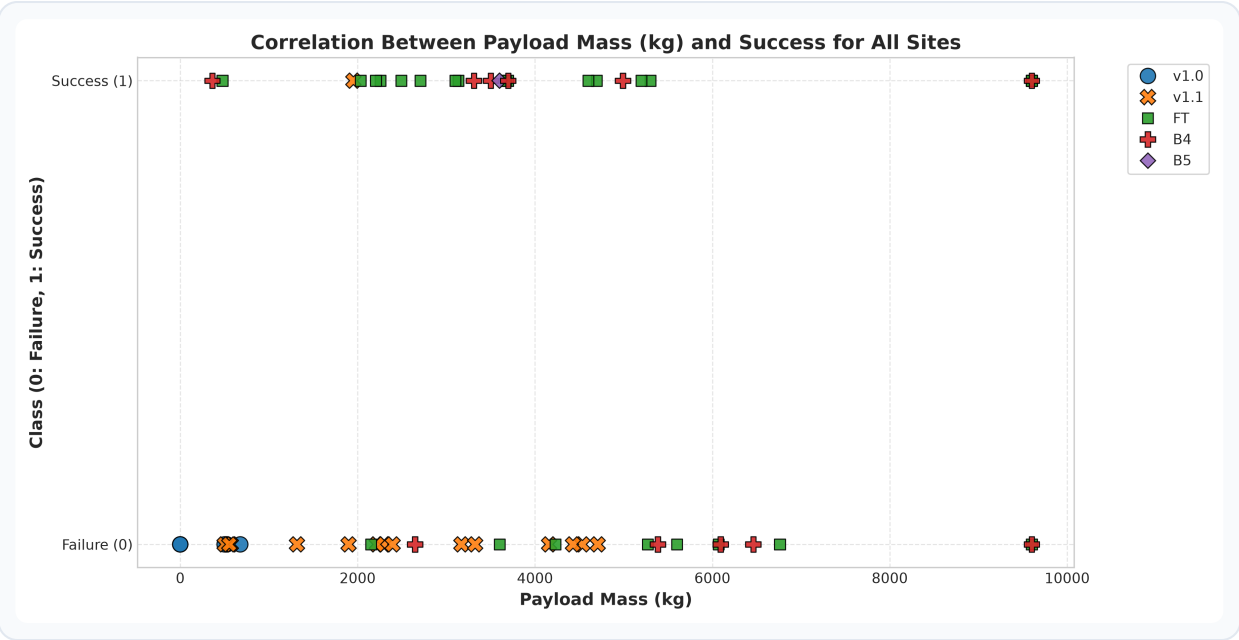
Launch Success Breakdown by Site



All Sites Breakdown: KSC LC-39A contributed the largest share of successful launches (**41.7%**), followed by CCAFS SLC-40 (**29.2%**).

Site Success Rate: Evaluating individual site ratios, **KSC LC-39A** demonstrated the highest overall landing success rate at **76.9%**.

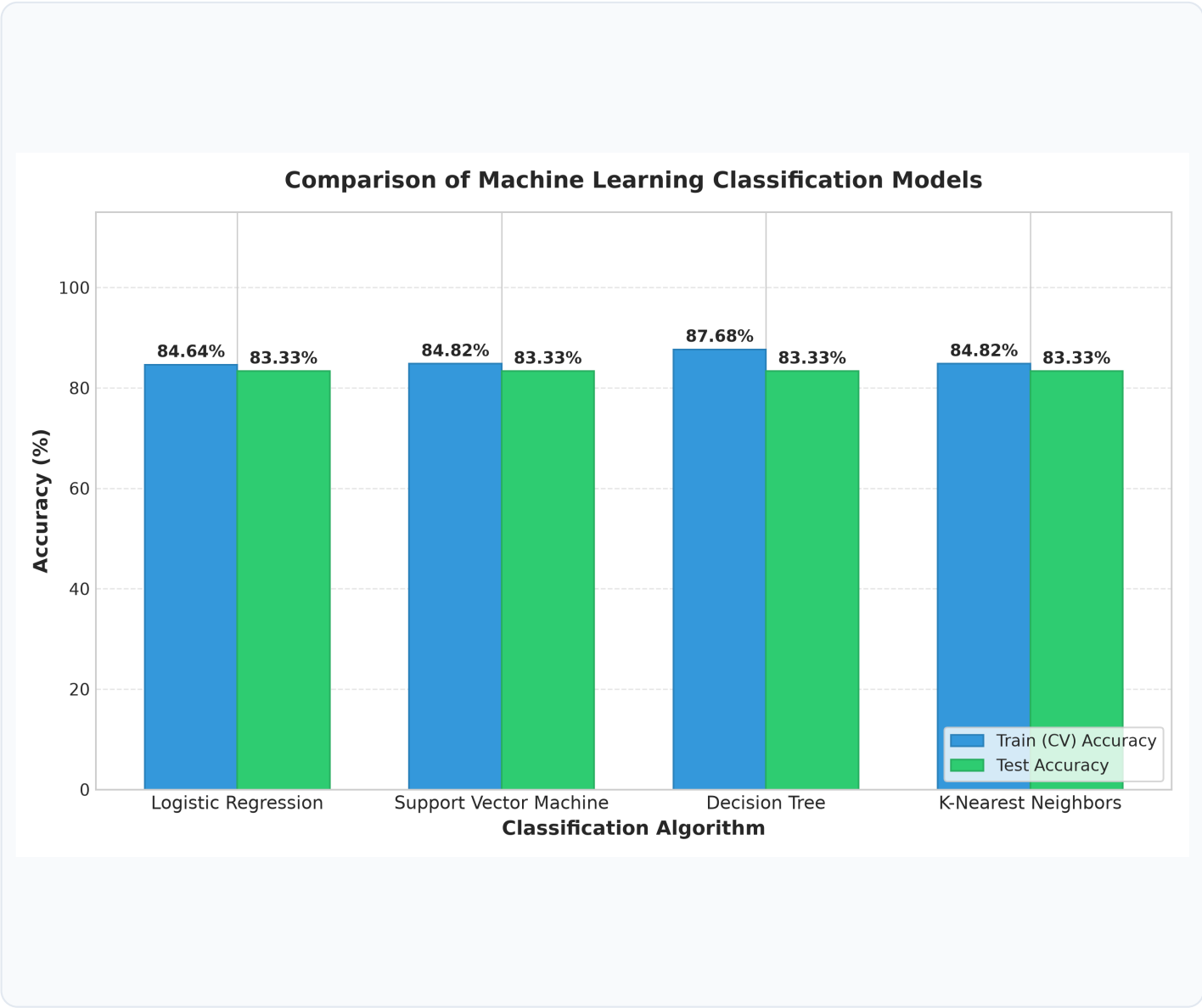
Interactive Scatter Analysis Answers



- **1. Highest Success Payload Range:** Payloads between **2,000 kg and 5,300 kg** exhibited the highest landing success rate.
- **2. Lowest Success Payload Range:** Payloads between **500 kg and 2,000 kg** (early v1.0 / v1.1 flights) had low recovery rates.
- **3. Booster Version Superiority:** The **FT (Full Thrust)** and **B5 (Block 5)** booster versions achieved near 100% success across all payload ranges.

Predictive Modeling: Model Accuracy Comparison

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>



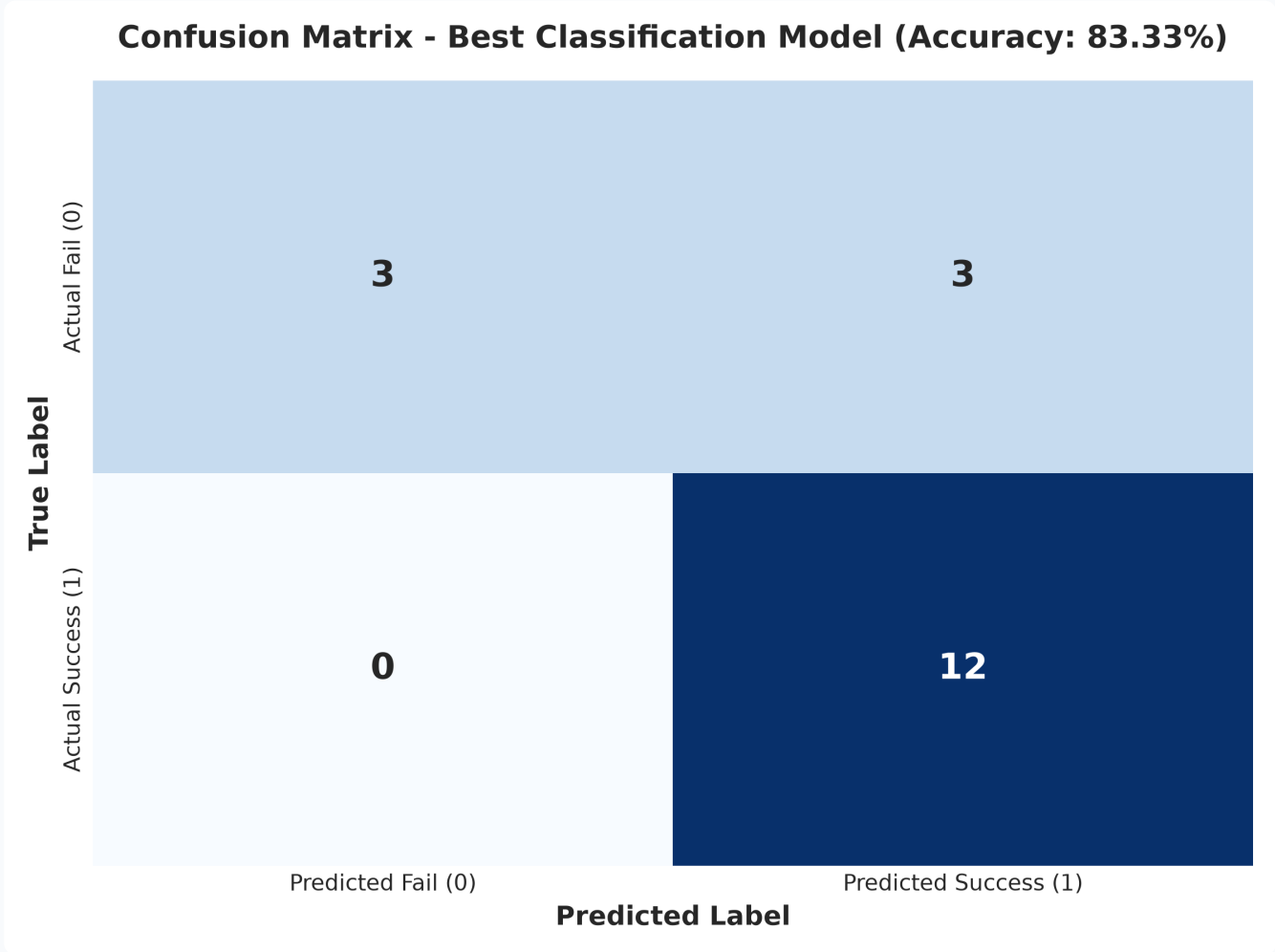
10-Fold CV & Test Evaluation Summary

Classification Model	10-Fold CV Score	Test Set Accuracy
Decision Tree Classifier	87.68%	83.33%
Support Vector Machine (SVM)	84.82%	83.33%
K-Nearest Neighbors (KNN)	84.82%	83.33%
Logistic Regression	84.64%	83.33%

Best Model Selection: The **Decision Tree Classifier** achieved the highest training cross-validation accuracy of **87.68%**, while all four algorithms achieved an identical test accuracy of **83.33%** on the 18 unseen test instances.

Predictive Modeling: Confusion Matrix Diagnostics

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>



Diagnostic Performance Analysis

- **True Positives (TP = 12):** The model accurately identified 12 successful landings out of 12 actual successes (**100% Sensitivity / Recall**).
- **True Negatives (TN = 3):** Successfully caught 3 landing failures.
- **False Positives (FP = 3):** 3 failed landings were predicted as successful.
- **False Negatives (FN = 0):** Zero successful landings were misclassified as failures.
- **Test Set Behavior:** Because the test set contains 18 samples, all models misclassified the same 3 challenging edge cases, yielding identical **83.33% test accuracy**.

Conclusions, Creativity Applied & Innovative Industry Insights

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

Key Scientific Conclusions & Creativity Applied

- **Predictive Viability:** Supervised classification models reliably forecast Falcon 9 booster recovery with **83.33% accuracy** based on pre-launch parameters.
- **Dominant Predictors:** Booster version (Block 5 superiority), target orbit inclination (LEO vs GTO delta-v requirement), and launch site infrastructure.
- **Learning Curve:** SpaceX's iterative engineering transitioned rocket recovery from a speculative experiment in 2010 to an industrialized utility by 2020.
- **Creativity Applied Beyond Template:** Built an end-to-end automated pipeline integrating REST APIs, web scrapers, SQLite, interactive GIS maps, a reactive Plotly Dash web server, and 300 DPI publication charts with unified executive color grading.

Innovative Insights & Business Recommendations

- **Competitive Bidding Model:** Competing launch providers (e.g., Rocket Lab, Blue Origin, Arianespace) can integrate this model to dynamically predict whether a given SpaceX bid entails reusable booster economics (\$62M) or expendable margins (\$90M+).
- **Launch Insurance Hedging:** Satellite operators can leverage predictive landing analytics to negotiate secondary payload insurance premiums and orbital insertion risk discounts.
- **Operational Extension:** Adding real-time sea-state (wave height) and upper-atmosphere wind telemetry could push predictive accuracy past 90%.

Repository Verification & Code Artifacts

GitHub: <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>

GitHub Repository Directory Structure

```
SpaceX-Falcon9-Data-Science-Capstone/
├── README.md                # Complete project documentation
├── data/                    # All cleaned datasets & SQLite db
├── figures/                 # All high-res 300 DPI charts
├── notebooks/              # All 7 executed Jupyter Notebooks
│   ├── 1_Data_Collection_API.ipynb
│   ├── 2_Data_Collection_Web_Scraping.ipynb
│   ├── 3_Data_Wrangling.ipynb
│   ├── 4_EDA_with_SQL.ipynb
│   ├── 5_EDA_with_Visualization.ipynb
│   ├── 6_Interactive_Visual_Analytics_Folium.ipynb
│   └── 7_Machine_Learning_Prediction.ipynb
├── reports/
│   └── Data Science Capstone Project Report.pdf
├── scripts/
│   ├── spacex_dash_app.py    # Plotly Dash web dashboard
│   └── generate_analysis_and_figures.py
```

Evaluation Checklist Verification

- **GitHub Repository URL:** <https://github.com/Phantom-Pain9/SpaceX-Falcon9-Data-Science-Capstone>
- **Completed Notebooks:** All 7 notebooks executed with active outputs.
- **SQL & Visual EDA:** All 10 queries executed with tabular results.
- **Geospatial & Dash:** Folium maps rendered and Dash callbacks verified.
- **Machine Learning:** 4 algorithms tuned with GridSearchCV and Confusion Matrix.
- **Author:** Phantom Pain (@Phantom-Pain9)